BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 224 · Trees and binary search trees: implementation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a null Next mean in a singly linked list?

Q2. In a BST, where does a smaller key go?

Q3. What is the base case of a recursive insertion reaching an empty child?

Q4. Insert 10,20,30 in that order into an empty unbalanced BST. What shape results?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

binary search tree — Binary tree in which lower keys occur to the left and higher to the right.

root — Topmost tree node.

leaf — A node with no children.

in-order traversal — Visit left subtree, node, then right subtree.

successor — Next key in in-order sorted sequence.

New learning

Find the starting node — The root is 40, so compare 25 with 40 and move left.

Repeat comparisons — 25>20 goes right to 30; 25<30 goes left.

Insert at a null child — Create node 25 at the empty left-child position of 30. This implementation ignores duplicate keys: when key equals node.Key, neither subtree changes.

Worked example

Insert 12 into the BST 20 (root), 10 (left), 30 (right).

1. Predict first — Compare 12 with 20, then 10.

2. Model the reasoning — Insert 12 as the right child of 10, preserving the BST property.

We Do · Practise with guidance

Insert 35 into BST root 25, left 15 and right 40.

```
35 > 25 → right
35 < 40 → left
```

Predict first

35>25 leads right from the root.

Model the reasoning

35<40 leads left, placing 35 as the left child of 40.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Insert 12 into BST root 15, left 9 and right 20. Identify its parent and position.

Question 2

Insert 50,30,70,20,40,35 into an empty BST. State parent and side of 35, and the in-order visit sequence.

Question 3

In BST 40 with children 20 and 60, and 20 with children 10 and 30, delete leaf 10. Explain a second case in which deletion is more involved.

Exit Ticket · Answer before checking

In-order traversal of a BST with distinct keys visits values in ascending order.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It marks the end of the list.

Q2. Into the left subtree.

Q3. Create and return a new node at that null position.

Q4. A right-leaning chain 10→20→30.

You Do pages · Independent questions

Q1. [4] 12<15 [1], moves left to 9 [1], 12>9 [1], so it is right child of 9 [1].

Q2. [6] Parent of 35 is 40 [1], it is left child [1]; in-order sequence 20,30,35,40,50,70 [4].

Q3. [6] Find node 10 by moving left twice [2]; it is a leaf [1], so set parent 20.Left=null [1]. For a node with two children, replace it using an in-order successor/predecessor and remove the replacement from its old location [2].

Exit Ticket · Model answer

True. Left subtree, node and right subtree produce increasing keys.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: